



```
In [2]: import spacy
        from fastcoref import FCoref
        from collections import defaultdict
        import json
```

```
c:\Python314\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IPProgress not found.
Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

```
In [9]: # file = "DATA/BASIL-main/articles/2019/0d10341a-9dba-4374-a524-814c300d1611_1
# file = "DATA/BASIL-main/articles/2016/7a97de89-1433-46f7-bcc2-f80b041cb9a0_2
# file = "DATA/BASIL-main/articles/2016/d4a4bc34-e48b-4485-86b2-f1daf4d464dd_1
# file = "DATA/BASIL-main/articles/2015/2b9468af-4ab0-4f25-85f4-c3ef93d72006_1
file = "DATA/BASIL-main/articles/2016/59f2f47c-5da0-4fbc-be6e-4f15d45d82e2_3.j
def to_article_BASIL(filename: str) -> list[str]:
    ret: list[str] = []
    with open(filename, "r", encoding="utf-8") as f:
        data = json.load(f)

    #Export from nested list to a single list of sentences
    for paragraph in data["body-paragraphs"]:
        if len(paragraph) == 1:
            ret.append(paragraph[0].lower() + " ")
        else:
            for sentence in paragraph:
                ret.append(sentence.lower() + " ")

    return ret
article = []
Sentence_list = to_article_BASIL(file)
#get only the first ten sentences
text = "".join(Sentence_list).lower()
text
```

Out[9]: "sen. ted cruz has laid down the gauntlet in the growing war of words with donald trump over the republican presidential candidates' wives. the texas lawmaker blamed his rival for the national enquirer story published friday that alleged cruz has had extramarital affairs, and after hits from trump's campaign about the mental health of his wife, the firebrand senator appears to be backing away from his pledge to support trump if the billionaire real estate mogul wins the gop nomination. "i don't make a habit out of supporting people who attack my wife and attack my family," cruz said at a press conference after a wisconsin rally, according to the hill. he added, "donald trump is not going to be the republican nominee." earlier in the campaign, gop candidates signed a promise not to run as an independent if they were not nominated. the move was meant to convince voters that trump would remain faithful to the republican party even if he lost. "it is garbage, complete and utter lies," cruz said about the national enquirer article alleging that he has had multiple affairs. "it is a tabloid smear, and it has come from donald trump and his henchmen." trump spokeswoman katrina pierson responded to claims that she is one of the mystery women referred to in the article, stating on twitter that the allegation is "stupidity on all levels." "of course the national enquirer story is 100 percent false!," pierson tweeted. of course the national enquirer story is 100% false!!! i only speak to myself, however. earlier this year, the tabloid endorsed trump for president with an issue that included an article giving ten reasons why the reality tv star is the "only choice for the white house." the intensely personal feud between cruz and trump began earlier this week when a super pac that opposes trump posted a meme-style ad featuring a naked melania trump that trump's then-girlfriend, a professional model, posed for more than 15 years ago for british gq. make america awesome pac paid for the ads and posted them on facebook and instagram. the photo features mrs. trump sprawled out naked on a fur rug inside a boeing 727. the social media posts read, "meet melania trump. your next first lady. or you could support ted cruz on tuesday." an angry trump lashed out at cruz on twitter, calling the senator a liar, as he has alleged before in the campaign, and threatening to "spill the beans" about cruz's wife heidi. "lyin' ted cruz just used a picture of melania from a gq shoot in his ad," trump said on the social media site. "be careful, lyin' ted, or i will spill the beans on your wife!" cruz quickly retorted by saying that the ads did not come from his campaign, and called trump a "coward" threatening to go after his wife. "pic of your wife not from us," cruz said. "donald, if you try to attack heidi, you're more of a coward than i thought. #classless." the lurch into the sort of personal territory that's normally off limits in campaigns occurred as an anti-trump super pac began running an ad in primary states featuring women reciting comments made by the billionaire that are derogatory toward women. that ad was produced by our principles, a group founded by a former mitt romney campaign adviser who is trying to help the gop appeal to women. trump has a substantial lead in the delegate chase for the gop nomination. cruz has a stiff challenge trying to catch him in remaining races and may only have a shot at the nomination if the contest spills into the summer convention. "

```
In [10]: raw_text = ""
AUGUSTA, Me. — Paul R. LePage, the embattled Republican governor of Maine, declared
"I will not resign," said Mr. LePage, who tried to put to rest swirling questions
"I'm not an alcoholic, and I'm not a drug addict, and I don't have mental issues"
```

Mr. LePage nevertheless said he was seeking “spiritual guidance,” had apologized. Behind the scenes, state lawmakers – Democrats and Senate Republicans – exasperated. The controversy, which has galvanized the state, started last Wednesday when Mr. LePage said. Both Democrats and Republicans have denounced the remarks, as many legislators have. But on Tuesday, Mr. LePage’s Republican allies in the House, including the minority leader, supported the decision. That decision did not sit well with some other state Republicans, and on Wednesday, Representative Fredette said, “With all due respect, I completely disagree with Representative Fredette’s position.” Mr. Thibodeau said many Senate Republicans were still deciding whether or not to support the decision. “There should be consequences for our actions,” Mr. Thibodeau said, “and I think there should be.” There is no precedent for such a situation here – a fact that has sent lawmakers scrambling. “It’s never been done before, so we’re in a little bit of uncharted waters,” said a Democratic source. Democrats have continued to call for Mr. LePage’s ouster, and on Wednesday said they would vote to remove him. “Coming back in for a special session will be discussed and all options around Mr. LePage’s six years in office here, which began as part of the Tea Party swing, will be on the table.” By all indications, Mr. LePage seemed determined to again ride out the political storm. “I want to put this behind us,” Mr. LePage said, during a meeting with a select group of lawmakers. Mr. LePage spoke shortly after he met with – and apologized to – Representative Fredette. But he but did not apologize for statements he made last week that linked traffic deaths to race. “I didn’t use racially charged language,” Mr. LePage claimed, although he also said he was “sorry.”

```
In [10]: nlp = spacy.load("en_core_web_trf")
doc = nlp(text)
```

```
In [11]: for ent, chunk in zip(doc.ents, doc.noun_chunks):
print(ent.text, ent.label_, chunk.text)
```

ted cruz PERSON sen. ted cruz
donald trump PERSON the gauntlet
republican NORP the growing war
texas GPE words
national enquirer ORG donald trump
friday DATE the republican presidential candidates' wives
cruz PERSON the texas lawmaker
trump PERSON his rival
trump PERSON the national enquirer story
gop ORG that
cruz PERSON cruz
wisconsin GPE extramarital affairs
hill ORG hits
donald trump PERSON trump's campaign
republican NORP the mental health
gop ORG his wife
trump PERSON the firebrand senator
the republican party ORG his pledge
cruz PERSON trump
national enquirer ORG the billionaire real estate mogul
donald trump PERSON the gop nomination
trump PERSON i
katrina pierson PERSON a habit
twitter ORG people
100 percent PERCENT who
pierson PERSON my wife
national enquirer ORG my family
100% PERCENT cruz
earlier this year DATE a press conference
trump PERSON a wisconsin rally
ten CARDINAL the hill
the white house ORG he
cruz ORG donald trump
trump PERSON the republican nominee
earlier this week DATE the campaign
trump PERSON gop candidates
melania trump PERSON a promise
trump PERSON an independent
more that 15 years ago DATE they
british gq ORG the move
america awesome pac ORG voters
facebook ORG trump
instagram ORG the republican party
trump PERSON he
boeing ORG it
727 PRODUCT garbage
melania trump PERSON complete and utter lies
ted cruz PERSON cruz
tuesday DATE the national enquirer article
trump PERSON he
cruz PERSON multiple affairs
cruz PERSON it
heidi PERSON a tabloid smear
lyin' ted cruz PERSON it

```

melania PERSON donald trump
gq ORG his henchmen
trump PERSON trump spokeswoman katrina pierson
lyin' PERSON claims
ted PERSON she
cruz PERSON the mystery women
trump PERSON the article
cruz PERSON twitter
donald PERSON the allegation
heidi PERSON stupidity
our principles ORG all levels
mitt romney PERSON the national enquirer story
gop ORG pierson
trump PERSON the national enquirer story
gop ORG i
cruz PERSON myself
summer DATE the tabloid

```

```
In [12]: Name_TAG = {(ent.text, ent.label_) for ent in doc.ents if ent.label_ == "PERSON"}
```

```
In [13]: Name_TAG
```

```
Out[13]: {('cruz', 'PERSON'),
          ('donald', 'PERSON'),
          ('donald trump', 'PERSON'),
          ('heidi', 'PERSON'),
          ('katrina pierson', 'PERSON'),
          ('lyin'', 'PERSON'),
          ('lyin' ted cruz', 'PERSON'),
          ('melania', 'PERSON'),
          ('melania trump', 'PERSON'),
          ('mitt romney', 'PERSON'),
          ('pierson', 'PERSON'),
          ('republican', 'NORP'),
          ('ted', 'PERSON'),
          ('ted cruz', 'PERSON'),
          ('texas', 'GPE'),
          ('trump', 'PERSON'),
          ('wisconsin', 'GPE')}
```

```
In [14]: Name_TAG = {(ent.text, ent.label_) for ent in doc.ents if ent.label_ == "PERSON"}
# NAME_ONLY = {ent.text for ent in doc.ents if ent.label_ == "PERSON" or ent.label_ == "NORP"}
lastname = {(n.split()[-1], tag) for n,tag in Name_TAG}
lastname
```

```
Out[14]: {('cruz', 'PERSON'),
          ('donald', 'PERSON'),
          ('heidi', 'PERSON'),
          ('lyin', 'PERSON'),
          ('melania', 'PERSON'),
          ('pierson', 'PERSON'),
          ('republican', 'NORP'),
          ('romney', 'PERSON'),
          ('ted', 'PERSON'),
          ('texas', 'GPE'),
          ('trump', 'PERSON'),
          ('wisconsin', 'GPE')}
```

```
In [7]: model = FCoref(device='cuda:0')
```

```
04/12/2026 11:42:33 - INFO - missing_keys: []
04/12/2026 11:42:33 - INFO - unexpected_keys: []
04/12/2026 11:42:33 - INFO - mismatched_keys: []
04/12/2026 11:42:33 - INFO - error_msgs: []
04/12/2026 11:42:33 - INFO - Model Parameters: 90.5M, Transformer: 8
2.1M, Coref head: 8.4M
```

```
In [8]: preds = model.predict(texts=text)
```

```
04/12/2026 11:42:36 - INFO - Tokenize 1 inputs...
Map: 100%|██████████| 1/1 [00:00<00:00, 40.26 examples/s]
04/12/2026 11:42:36 - INFO - ***** Running Inference on 1 texts *****
Inference: 100%|██████████| 1/1 [00:00<00:00, 1.98it/s]
```

```
In [78]: cluster = preds.get_clusters()
spans: list[str] = preds.get_clusters(as_strings=False)

pronouns = {
    "i", "me", "my", "mine", "myself", "we", "us", "our", "ours", "ourselves",
    "you", "your", "yours", "yourself", "yourselves", "he", "him", "his", "him",
    "she", "her", "hers", "herself", "it", "its", "itself", "they", "them",
    "their", "theirs", "themselves", "who", "whom", "whose", "which", "what",
    "this", "that", "these", "those", "anyone", "someone", "no one", "everyone"
}

def create_dictionary( cluster, spans):
    my_dict = defaultdict(list)
    for nouns, indexes in zip(cluster, spans):
        for noun in nouns:
            if noun not in pronouns:
                key = noun
                break
            my_dict[key].extend(indexes)

    return my_dict
```

```
my_dict = create_dictionary(cluster, spans)
my_dict.keys()
```

```
Out[78]: dict_keys(['donald j. trump', 'mexico', 'his plan to build a giant wall along
the border with mexico', 'a giant wall', 'vicente fox', 'thursday', 'hispanic
voters', '"mexican people', '"mr. fox's successor, felipe calderón,', '2006',
'the border with mexico'])
```

```
In [83]: Name_TAG
```

```
Out[83]: {'america',
'calderón',
'donald j. trump',
'felipe calderón',
'fox',
'hispanic',
'hispanics',
'mexican',
'mexico',
'the united states',
'trump',
'vicente fox'}
```

```
In [86]: string_name = []
for name in Name_TAG:
    string_name.extend(name.split())
newset = set(string_name)
newset
```

```
Out[86]: {'america',
'calderón',
'donald',
'felipe',
'fox',
'hispanic',
'hispanics',
'j.',
'mexican',
'mexico',
'states',
'the',
'trump',
'united',
'vicente'}
```

```
In [75]: newlist = list(my_dict.keys())
for name in newlist:
    name.split()
```

```
In [ ]: new_dict = defaultdict(list)
```

```
for key, value in list(my_dict.items()):
```

[illegible]

```
In [62]: for k,v in my_dict.items():
          print(k)
```

donald j. trump
mexico
his plan to build a giant wall along the border with mexico
a giant wall
vicente fox
thursday
hispanic voters
"mexican people
"mr. fox's successor, felipe calderón,
2006
the border with mexico

```
In [63]: for k,v in new_dict.items():
```

```
print(k)
```

```
donald j. trump  
mexico  
his plan to build a giant wall along the border with mexico  
a giant wall  
vicente fox  
thursday  
hispanic voters  
"mexican people  
"mr. fox's successor, felipe calderón,  
2006  
the border with mexico
```

```
In [49]: for name, tag in Name_TAG:  
         print(name)
```

```
trump  
felipe calderón  
america  
calderón  
hispanics  
vicente fox  
the united states  
mexico  
fox  
mexican  
hispanic  
donald j. trump
```